

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	Ramana Y	Reg. No.:	192525476
Date:		Duration:	60 minutes

Code of Conduct

I Ramana Y (192525476) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Tvest Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

1. Introduction

FoodRescue Connect is a smart food donation and redistribution management system designed to connect food donors, NGOs, volunteers, and beneficiaries. The system manages food donations, requests, allocation, pickup, delivery, notifications, tracking, and analytical reporting.

The objective of this test case design is to verify the **correctness, functionality, reliability, security, performance, and usability** of the proposed FoodRescue Connect system.

The test cases cover:

- Normal conditions
- Boundary conditions
- Invalid inputs
- Exceptional conditions
- Functional requirements
- Non-functional requirements
- Equivalence Partitioning
- Boundary Value Analysis
- Decision Table Testing
- State Transition Testing
- Requirement traceability
- Test coverage

2. Functional Requirements to be Tested

Requirement ID	Functional Requirement
FR-01	User registration and login
FR-02	Role-based access for Donor, NGO, Volunteer, Beneficiary, and Admin
FR-03	Donor can create a food donation
FR-04	Donor can update or cancel a donation
FR-05	System records food quantity and expiry information
FR-06	NGO can search and request available donations
FR-07	System allocates food based on availability and priority
FR-08	AI module predicts food priority/shelf-life and recommends allocation
FR-09	Route optimization identifies suitable volunteers and delivery routes
FR-10	Volunteers can accept pickup and delivery requests
FR-11	System tracks pickup and delivery status
FR-12	System sends notifications for donations, requests, pickups, and deliveries
FR-13	Beneficiaries can receive allocated food
FR-14	Admin can monitor users and donation activities
FR-15	System generates donation and waste-reduction reports
FR-16	System integrates with GPS/maps and external notification services

Requirement ID	Functional Requirement
FR-17	Food quality information can be monitored
FR-18	System maintains donation history and status

3. Non-Functional Requirements to be Tested

Requirement ID	Non-Functional Requirement	Testing Focus
NFR-01	Security	Authentication, authorization, data protection
NFR-02	Performance	Response time and concurrent users
NFR-03	Availability	System availability and recovery
NFR-04	Reliability	Correct processing and failure recovery
NFR-05	Scalability	Increasing users and donations
NFR-06	Usability	Ease of navigation and error messages
NFR-07	Maintainability	Modular functionality and service isolation
NFR-08	Interoperability	Maps, GPS, notification and external APIs

4. Test Scenarios

User Management

- Successful user registration
- Duplicate email registration
- Invalid email
- Weak password
- Successful login
- Incorrect password
- Unauthorized role access
- Logout

Donation Management

- Create valid donation
- Missing donation information
- Zero quantity
- Negative quantity
- Expired food
- Donation cancellation
- Donation update
- Large quantity donation

Food Request and Allocation

- NGO requests available food
- NGO requests unavailable food
- Multiple NGOs request the same donation
- Priority-based allocation

- Request cancellation
- Allocation after food expiry

AI Recommendation

- Shelf-life prediction
- Urgent donation prioritization
- NGO matching
- Insufficient prediction data
- AI service unavailable

Volunteer and Delivery

- Volunteer accepts pickup
- Volunteer rejects pickup
- Nearest volunteer selection
- Route calculation
- GPS unavailable
- Pickup confirmation
- Delivery confirmation

Notifications

- New donation notification
- Pickup notification
- Delivery notification
- Expiry notification
- Notification service failure

Tracking

- Live location update
- Pickup status update
- Delivery status update
- Invalid status transition
- GPS interruption

Reporting

- Donation report generation
- Waste reduction report
- Empty report period
- Large report generation

5. Detailed Test Cases

A. User Registration and Authentication

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-001	Valid registration	Open registration → enter valid details → submit	Name: Ravi, Email: ravi@gmail.com , Password: Ravi@123	Account should be created successfully	Not Executed	Not Executed
TC-002	Duplicate registration	Register using an existing email	ravi@gmail.com	System should reject duplicate email	Not Executed	Not Executed
TC-003	Invalid email	Enter invalid email and submit	ravi@	Validation error should be displayed	Not Executed	Not Executed
TC-004	Empty mandatory fields	Leave required fields empty	Name = blank	System should display required-field errors	Not Executed	Not Executed
TC-005	Weak password	Enter password below required strength	12345	System should reject weak password	Not Executed	Not Executed
TC-006	Valid login	Enter correct credentials	ravi@gmail.com / Ravi@123	User should be successfully logged in	Not Executed	Not Executed
TC-007	Incorrect password	Enter correct email and wrong password	ravi@gmail.com / wrong123	Login should fail with an appropriate error	Not Executed	Not Executed
TC-008	Unauthorized access	Login as beneficiary and access admin page	Beneficiary account	Access should be denied	Not Executed	Not Executed

B. Food Donation Management

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-009	Create valid donation	Login as donor → Add Donation → submit	Rice, 50 kg, expiry tomorrow	Donation should be created successfully	Not Executed	Not Executed
TC-010	Missing food name	Leave food name empty	Quantity = 10 kg	System should display validation error	Not Executed	Not Executed
TC-011	Zero quantity	Enter zero quantity	Quantity = 0 kg	System should reject donation	Not Executed	Not Executed

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-012	Negative quantity	Enter negative quantity	Quantity = -5 kg	System should reject input	Not Executed	Not Executed
TC-013	Valid expiry date	Enter future expiry date	Tomorrow	Donation should be accepted	Not Executed	Not Executed
TC-014	Expired food	Enter past expiry date	Yesterday	System should reject expired food	Not Executed	Not Executed
TC-015	Update donation	Edit existing donation	Quantity 50 → 40 kg	Updated quantity should be stored	Not Executed	Not Executed
TC-016	Cancel donation	Select active donation → Cancel	Donation ID D001	Donation status should change to Cancelled	Not Executed	Not Executed
TC-017	Large quantity	Enter unusually large but valid quantity	10,000 kg	System should process or provide appropriate validation	Not Executed	Not Executed

C. NGO Food Request and Allocation

a	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-018	Request available food	NGO searches available donation → Request	D001, 20 kg	Request should be created	Not Executed	Not Executed
TC-019	Request unavailable food	Request an already allocated donation	D002	System should reject request	Not Executed	Not Executed
TC-020	Request greater quantity	Request more than available quantity	Available 20 kg, Request 30 kg	System should reject or partially allocate according to business rules	Not Executed	Not Executed
TC-021	Multiple NGO requests	Two NGOs request same donation	NGO-A and NGO-B	System should allocate according to priority/business rules without double allocation	Not Executed	Not Executed
TC-022	Cancel request	NGO cancels pending request	Request R001	Request should become Cancelled	Not Executed	Not Executed
TC-023	Expired donation request	Request food after expiry	Expired donation	Request should not be accepted	Not Executed	Not Executed

D. AI Recommendation

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-024	Food priority prediction	Submit donation details to AI module	Expiry in 2 hours	Donation should receive high urgency	Not Executed	Not Executed
TC-025	Normal priority prediction	Submit donation with sufficient shelf life	Expiry in 2 days	Donation should receive normal priority	Not Executed	Not Executed
TC-026	NGO recommendation	Run matching process	Donation + nearby NGOs	Suitable NGO should be recommended	Not Executed	Not Executed
TC-027	Missing AI input	Submit incomplete food information	Missing expiry	System should handle prediction failure gracefully	Not Executed	Not Executed
TC-028	AI service unavailable	Stop AI service → request recommendation	Valid donation	System should display fallback/error and continue core operations	Not Executed	Not Executed

E. Volunteer and Route Optimization

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-029	Volunteer accepts pickup	Volunteer opens request → Accept	Pickup P001	Pickup should be assigned to volunteer	Not Executed	Not Executed
TC-030	Volunteer rejects pickup	Volunteer selects Reject	Pickup P002	Pickup should remain available for reassignment	Not Executed	Not Executed
TC-031	Nearest volunteer selection	Run route optimization	3 volunteers at different distances	Nearest suitable volunteer should be selected	Not Executed	Not Executed
TC-032	Route calculation	Enter donor and NGO locations	Location A → Location B	Optimized route should be displayed	Not Executed	Not Executed
TC-033	GPS unavailable	Disable GPS during route calculation	GPS unavailable	System should show an appropriate error/fallback	Not Executed	Not Executed
TC-034	Pickup confirmation	Volunteer confirms collection	P001	Status should change to Picked Up	Not Executed	Not Executed
TC-035	Delivery confirmation	Volunteer confirms delivery	D001	Status should change to Delivered	Not Executed	Not Executed

F. Notification and Tracking

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-036	New donation notification	Create donation	D001	Relevant NGOs should receive notification	Not Executed	Not Executed
TC-037	Pickup notification	Assign volunteer	P001	Volunteer should receive pickup notification	Not Executed	Not Executed
TC-038	Delivery notification	Mark donation delivered	D001	Donor/NGO should receive delivery notification	Not Executed	Not Executed
TC-039	Expiry reminder	Donation approaches expiry	1 hour remaining	System should send expiry alert	Not Executed	Not Executed
TC-040	Notification service failure	Disable notification service	Valid event	Event should be retained/retried without data loss	Not Executed	Not Executed
TC-041	Live tracking	Volunteer moves during delivery	GPS coordinates	Location should update correctly	Not Executed	Not Executed
TC-042	Invalid status transition	Change Delivered back to Pending	Delivered → Pending	Invalid transition should be rejected	Not Executed	Not Executed

G. Reporting and Analytics

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-043	Generate donation report	Select date range → Generate	01/08/2026–15/08/2026	Correct donation statistics should be displayed	Not Executed	Not Executed
TC-044	Waste reduction report	Open analytics dashboard	Existing donation data	Waste reduction statistics should be calculated correctly	Not Executed	Not Executed
TC-045	Empty date range	Generate report for period with no donations	No data	System should display an appropriate empty-state message	Not Executed	Not Executed
TC-046	Large report	Generate report containing large dataset	100,000 records	Report should generate without system failure	Not Executed	Not Executed

6. Non-Functional Test Cases

TC ID	Attribute	Test Scenario	Expected Result
NTC-001	Performance	Send normal API request	Response should be returned within defined response-time target

TC ID	Attribute	Test Scenario	Expected Result
NTC-002	Performance	Simulate many concurrent users	System should continue responding without unacceptable degradation
NTC-003	Security	Attempt login with invalid credentials repeatedly	System should prevent unauthorized access
NTC-004	Security	Access admin API using beneficiary token	Request should be rejected
NTC-005	Security	Attempt SQL injection through login fields	Malicious input should not execute
NTC-006	Security	Access application using HTTP	System should redirect/block insecure communication according to deployment policy
NTC-007	Reliability	Database connection temporarily fails	System should handle failure without corrupting transactions
NTC-008	Availability	Restart one service	Other services should remain operational where independent
NTC-009	Scalability	Increase number of donations significantly	System should continue processing without architectural failure
NTC-010	Usability	New user performs donation creation	User should be able to complete task without unclear navigation
NTC-011	Interoperability	Request Google Maps route	Correct route information should be received
NTC-012	Recovery	Notification service becomes unavailable and recovers	Pending events should be processed after recovery

7. Boundary Value Analysis

Boundary Value Analysis is applied to inputs where values at or around limits can cause failures.

Example: Food Quantity

Assume the system accepts quantities greater than 0 kg.

Test ID	Quantity	Category	Expected Result
BVA-01	-1 kg	Below boundary	Reject
BVA-02	0 kg	Boundary	Reject
BVA-03	0.01 kg	Just above boundary	Accept
BVA-04	1 kg	Valid value	Accept
BVA-05	Maximum allowed quantity	Upper boundary	Accept
BVA-06	Maximum + 1	Above boundary	Reject

Example: Expiry

Test ID	Expiry	Expected Result
BVA-07	Already expired	Reject
BVA-08	Expiry exactly at current time	Reject/flag according to business rule
BVA-09	Expiry shortly in future	Accept with urgent priority
BVA-10	Normal future expiry	Accept

8. Equivalence Partitioning

The input domain is divided into valid and invalid classes.

Food Quantity

Partition	Example	Expected
EP-01 Valid positive quantity	10 kg	Accept
EP-02 Zero quantity	0 kg	Reject
EP-03 Negative quantity	-10 kg	Reject
EP-04 Invalid characters	"abc kg"	Reject
EP-05 Excessive quantity	Above maximum	Reject/validate

Email

a	Example	Expected
EP-06 Valid email	user@gmail.com	Accept
EP-07 Missing @	usergmail.com	Reject
EP-08 Missing domain	user@	Reject
EP-09 Empty	blank	Reject

9. Decision Table Testing

Donation Allocation Decision Table

Conditions:

- Food available?
- Food valid/not expired?
- NGO request available?
- Request quantity within available quantity?
- Priority level?

Rule	Food Available	Valid Food	Request Exists	Quantity Available	Priority	Action
R1	Yes	Yes	Yes	Yes	High	Allocate immediately
R2	Yes	Yes	Yes	Yes	Normal	Allocate normally
R3	Yes	Yes	Yes	No	High	Partial allocation / alternate donation
R4	Yes	No	Yes	Yes	High	Reject donation

Rule	Food Available	Valid Food	Request Exists	Quantity Available	Priority	Action
R5	No	-	Yes	-	High	Reject request
R6	Yes	Yes	No	Yes	-	Keep donation available
R7	Yes	Yes	Yes	Yes	Low	Queue/normal allocation

This ensures that food is not incorrectly allocated when it is unavailable or expired.

10. State Transition Testing

The donation lifecycle can be represented as:

Created → Available → Requested → Allocated → Pickup Assigned → Picked Up → Delivered

Alternative states:

Available → Cancelled

Requested → Rejected

Available → Expired

State Transition Test Cases

Test ID	Current State	Event	Expected Next State
ST-01	Created	Publish donation	Available
ST-02	Available	NGO requests food	Requested
ST-03	Requested	NGO accepts	Allocated
ST-04	Allocated	Volunteer assigned	Pickup Assigned
ST-05	Pickup Assigned	Food collected	Picked Up
ST-06	Picked Up	Food delivered	Delivered
ST-07	Available	Donor cancels	Cancelled
ST-08	Available	Expiry reached	Expired
ST-09	Delivered	Attempt to cancel	Reject invalid transition
ST-10	Delivered	Attempt to return to Available	Reject invalid transition

11. Exceptional Condition Testing

Test ID	Exceptional Condition	Expected System Behaviour
EX-01	Database unavailable	Display error and prevent data corruption
EX-02	AI service unavailable	Use fallback mechanism or display recommendation unavailable

Test ID	Exceptional Condition	Expected System Behaviour
EX-03	Maps API unavailable	Display route service unavailable and allow retry
EX-04	Notification service unavailable	Queue notification and retry
EX-05	Internet connection lost	Preserve unsent information where applicable
EX-06	GPS signal lost	Display last known location and indicate tracking interruption
EX-07	Concurrent NGO requests	Prevent double allocation using transaction control
EX-08	Session expires during donation	Require login again and prevent unauthorized submission
EX-09	Duplicate submission	Prevent duplicate donation creation
EX-10	Server restart during transaction	Recover safely without inconsistent donation status

12. Requirement Traceability Matrix

Requirement	Related Test Cases
FR-01 User Registration/Login	TC-001 to TC-007
FR-02 Role-Based Access	TC-008, NTC-004
FR-03 Create Donation	TC-009
FR-04 Update/Cancel Donation	TC-015, TC-016
FR-05 Quantity/Expiry	TC-011 to TC-014, BVA-01 to BVA-10
FR-06 NGO Request	TC-018 to TC-023
FR-07 Food Allocation	TC-021, Decision Table R1-R7
FR-08 AI Recommendation	TC-024 to TC-028
FR-09 Route Optimization	TC-031 to TC-033
FR-10 Volunteer Management	TC-029, TC-030
FR-11 Tracking	TC-034, TC-035, TC-041, TC-042
FR-12 Notifications	TC-036 to TC-040
FR-13 Beneficiary Distribution	TC-035
FR-14 Administration	TC-008, NTC-004
FR-15 Reporting	TC-043 to TC-046
FR-16 External Integration	TC-032, TC-033, NTC-011
FR-17 Quality Monitoring	TC-024, TC-027
FR-18 Donation History	TC-043, TC-044
NFR-01 Security	NTC-003 to NTC-006
NFR-02 Performance	NTC-001, NTC-002
NFR-03 Availability	NTC-008
NFR-04 Reliability	NTC-007, NTC-012
NFR-05 Scalability	NTC-009

Requirement	Related Test Cases
NFR-06 Usability	NTC-010
NFR-08 Interoperability	NTC-011, NTC-012

13. Test Coverage Summary

Testing Area	Number of Test Cases
User Management	8
Donation Management	9
NGO Request & Allocation	6
AI Recommendation	5
Volunteer & Route Optimization	7
Notification & Tracking	7
Analytics & Reporting	4
Non-Functional Testing	12
Boundary Value Analysis	10
Equivalence Partitioning	9
Decision Table	7 rules
State Transition Testing	10
Exceptional Conditions	10

The test design provides coverage across the major functional modules and the important non-functional characteristics of FoodRescue Connect.

14. Test Design Techniques Applied

1. Equivalence Partitioning

Used for:

- Email validation
- Food quantity
- User input fields

It reduces the number of test cases while maintaining representative input coverage.

2. Boundary Value Analysis

Used for:

- Minimum food quantity
- Maximum food quantity
- Expiry time
- Input length

It tests values immediately below, at, and above boundaries.

3. Decision Table Testing

Used for:

- Food allocation
- Availability
- Expiry
- NGO requests
- Priority handling

It ensures that combinations of conditions result in the correct system action.

4. State Transition Testing

Used for:

- Donation lifecycle
- Pickup lifecycle
- Delivery status

It verifies that the system permits only valid state transitions.

15. Conclusion

The proposed test case design provides comprehensive testing coverage for the FoodRescue Connect system. It verifies major functions including registration, food donation, NGO requests, AI-based recommendation, route optimization, volunteer management, tracking, notifications, and analytics.

The test design also covers security, performance, availability, reliability, scalability, usability, and interoperability. Boundary Value Analysis, Equivalence Partitioning, Decision Table Testing, and State Transition Testing are applied to improve test effectiveness.

The test cases can help identify incorrect inputs, invalid state transitions, service failures, security vulnerabilities, allocation conflicts, and integration problems before deployment. Therefore, the proposed testing approach supports the development of a reliable, secure, efficient, and user-friendly FoodRescue Connect platform.

Actual Result and Pass/Fail fields should be updated during actual execution of the test cases.

1. FUNCTIONAL REQUIREMENTS

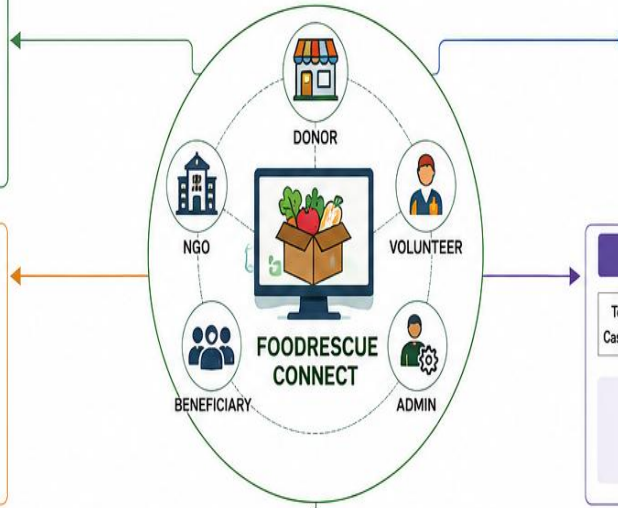
- User Registration & Login
- Role-Based Access Control
- Food Donation Management
- Food Request & Allocation
- AI Recommendation & Priority
- Route Optimization
- Volunteer Management
- Pickup & Delivery Tracking
- Notifications & Alerts
- Reporting & Analytics
- External Integrations (GPS, Maps, SMS, Email)
- Donation History & Quality Monitoring



FOODRESCUE CONNECT

Smart Food Donation and Redistribution Management System

TEST CASE DESIGN OVERVIEW



2. NON-FUNCTIONAL REQUIREMENTS

- Security
- Performance
- Reliability
- Availability
- Scalability
- Usability
- Maintainability
- Interoperability



3. TEST SCENARIOS

- User Management Scenarios
- Donation Management Scenarios
- Food Request & Allocation Scenarios
- AI Recommendation Scenarios
- Volunteer & Route Optimization Scenarios
- Notification & Tracking Scenarios
- Reporting & Analytics Scenarios
- Exceptional Scenarios



4. TEST CASE DESIGN

Test Case ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
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Covers Normal, Boundary, Invalid & Exceptional Conditions with detailed test data and expected results.

5. TEST DESIGN TECHNIQUES APPLIED

EQUIVALENCE PARTITIONING

Input Field	Valid Class	Invalid Class
Email	user@gmail.com	user@, user
Quantity	> 0	0, negative
Expiry Date	Future date	Past date
Food Name	Non-empty text	Blank, Special chars

Divides input domain into valid & invalid partitions and selects representative values.

BOUNDARY VALUE ANALYSIS

Field	Min	Min+1	Nominal	Max-1	Max
Quantity	-1	0.01	10	999	1000
Expiry (Days)	-1	0	2	30	31
Password Length	5	6	10	19	20

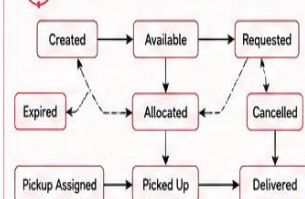
Tests values at the edges of boundaries to find defects at limit conditions.

DECISION TABLE TESTING

Conditions / Rules	R1	R2	R3	...	Rn
Food Available	Y	Y	N	...	Y
Food Valid	Y	N	Y	...	Y
Request Exists	Y	Y	Y	...	N
Quantity Available	Y	N	Y	...	Y
Priority High	Y	N	Y	...	N
Action	Allocate	Reject	Partial	...	Queue

Evaluates combinations of conditions and verifies correct system actions.

STATE TRANSITION TESTING



Validates correct state changes and prevents invalid transitions.

6. EXCEPTIONAL CONDITIONS

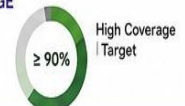
- Service Unavailable (AI/Maps/Notification)
- Database Failure
- Internet/GPS Unavailable
- Concurrent Requests
- Duplicate Submissions
- Session Timeout

7. NON-FUNCTIONAL TESTING

- Performance Testing
- Security Testing
- Usability Testing
- Reliability Testing
- Scalability Testing
- Compatibility Testing

8. TRACEABILITY & COVERAGE

- Requirement to Test Case Mapping
- Scenario to Test Case Mapping
- High Requirement Coverage



This Test Case Design ensures correctness, reliability, security, performance, and usability of the FoodRescue Connect System and supports quality assurance throughout the software development lifecycle.

